

Sarvesh Karandikar

Senior Software Engineer — Backend, Infra & AI

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SUMMARY

Senior Software Engineer with 5+ years at Cisco spanning backend infrastructure, cloud networking, platform security, and full-stack AI. Currently building multimodal agentic systems with the Claude Agent SDK and MCP — end to end, from model to platform to tooling. Fluent in both infrastructure (Go, Java, Kubernetes, AWS/GCP/Azure) and AI (LLMs, RAG, PyTorch, TensorFlow).

EXPERIENCE

Senior Software Engineer — AI Canvas

Oct 2025 — present

Cisco Systems

- ▶ Built multimodal Canvas Deep Agent — reasons over images, documents, and network captures across fast and plan-execute modes.
- ▶ Designed a secure file pipeline: gRPC → presigned S3 streaming (SSRF-guarded), Redis manifest store, MCP file-reader tooling.
- ▶ Own the Knowledge-Base SDK and skill/MCP-server registration powering self-serve partner onboarding.

Claude Agent SDK MCP Python gRPC Redis AWS

Software Engineer — Nexus Dashboard

Nov 2023 — Sep 2025

Cisco Systems

- ▶ Designed a Certificate Management System for automated CA creation and secure signing across Kubernetes communications.
- ▶ Enhanced external-CA integration to onboard Cisco XE/XR devices; built an agent-based Postgres/TSDb management interface.
- ▶ Implemented an SPI-based Java leader-election framework for active-standby configurations.

Java Go Kubernetes Postgres PKI

Software Engineer — Cloud Network Controller

Jul 2020 — Oct 2023

Cisco Systems

- ▶ Integrated GCP into the ACI model-based cloud-networking solution for multi-cloud compatibility and scale.
- ▶ Built config-drift security detection across L4-L7 service chains and a Terraform collection service feeding a unified graph database.
- ▶ Created a firewall-configuration analysis service using Binary Decision Diagrams to assess connectivity and traffic flows.

Python Terraform GCP AWS Azure Dgraph

WordPress Developer

2019 — 2020

USC Supply Chain

- ▶ Developed and maintained WordPress sites with custom themes and plugins for university supply-chain projects.
- ▶ Built REST API integrations connecting WordPress to backend inventory and logistics systems.
- ▶ Managed hosting, deployments, and performance optimization across multiple client sites.

WordPress

PHP

JavaScript

MySQL

REST APIs

ML Engineer Intern2019 — 2020

Circle Link (fintech)

- ▶ Built NER models classifying entities in loan documents with Elasticsearch indexing; fine-tuned OCR/MICR check recognition.

Python

NLP

Elasticsearch

OCR

SKILLS

Backend & Infrastructure

LANGUAGES

Go, Java, C++, Python, TypeScript, GraphQL

ORCHESTRATION & DATA

Docker, Kubernetes, Postgres, Redis, Dgraph

CLOUD & IAC

AWS, Azure, GCP, Terraform, OCI, Stacker

AI & Machine Learning

AGENTS & LLM

Claude Agent SDK, MCP, RAG, Amazon Bedrock

FRAMEWORKS & LIBRARIES

TensorFlow, PyTorch, NumPy, Pandas, NLTK

BRIDGES

Python, gRPC, Redis — the glue across both worlds

SELECTED PROJECTS

AlpacaBot

Weekly-options trading agent — Alpaca execution, Telegram control, terminal dashboard, AWS-hosted.

GitHub ↗

chromadb RAG

FastAPI ingest + semantic retrieval with centroid-based collection routing.

GitHub ↗

DriveSuite

WhatsApp → Jellyfin/Sonarr media bridge — local Llama for intent, Claude for execution.

GitHub ↗

EDUCATION

University of Southern California	MS, Computer Science	2020
SRM Institute of Science and Technology	BTech, Computer Science & Engineering	2019